

Introduction and Table of Contents



Commercial banks and insurance firms are the companies *most* different from traditional companies.

Within the Bank Modeling course, there are notes, slide summaries, and other guides, but this document summarizes **everything**.

So, you can use this guide to review everything right before interviews.

You can also use it to determine which parts of the course you need to study:

1. [How Banks & Insurance Firms Work](#)
2. [Bank Accounting 101](#)
3. [Insurance Accounting 101](#)
4. [Regulatory Capital](#)
5. [Valuation Multiples & Intrinsic Valuation](#)
6. [Bank-Specific and Insurance-Specific Valuation](#)
7. [M&A Deals and Merger Models](#)
8. [Buyout and Growth Equity Models](#)
9. [Recap and Summary](#)

How Banks & Insurance Firms Work

“Normal companies” sell products and services to customers. Customers pay them for these products and services, and companies record that money as revenue.

Commercial banks, by contrast, do *not* earn money by selling products directly to customers. Instead, they earn money by fulfilling two needs:

- 1) People need to **save** their money somewhere.
- 2) Other people need to **borrow** money when they don’t have enough.

Banks protect peoples’ money and incentivize people to store it with them by paying **interest** on it. This “stored money” is known as the bank’s **Deposits**.

Banks then pool that money together and lend it out in larger amounts to borrowers; the bank’s “lent-out-money” is known as **Loans**. The borrowers *pay the bank interest*.

Banks make money based on the **interest rate spread**: The difference between the Interest Rate on Loans and the Interest Rate on Deposits (and any other items that earn or bear interest).





Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

For example, you deposit \$1,000 from your paycheck into your JP Morgan bank account. JPM pays you a 1% interest rate on it.

Then, JPM takes that money you just deposited, puts it together with funds from thousands of other customers, and turns it into a \$100 million loan for a large company.

That large company pays JPM 4% interest on the loan, so the **interest rate spread** is $4\% - 1\% = 3\%$.

That number tells us how much in Net Interest Income the bank can earn as it grows its Loans and Deposits. For example, if the bank's Loans and Deposits both grow by \$100 million, it should earn $3\% * \$100 \text{ million} = \3 million in additional Net Interest Income.



Insurance firms are a bit different: Similar to traditional companies, they *do* sell their products directly to customers.

Specifically, they sell **policies** to customers, which require the customers to pay monthly **Premiums** in exchange for a payout when disaster strikes.

For example, an auto insurance policy might be: "Pay us \$100 / month, and if you get in a car accident, we'll pay for up to \$10,000 in repair costs"). Other types of policies could cover homes, health, or lives.

If you get in a car accident, the insurance company must pay you a **Claim** for \$10,000.

The insurance business model is more direct than the commercial bank business model, but many insurance firms, like banks, earn significant Interest Income.

That happens because insurance companies **collect Premiums upfront from customers, but they do not have to pay out anything to customers immediately.**

So, the Premiums act like a bank's Deposits – insurance firms can invest the customers' payments in stocks, bonds, real estate, and other asset classes until they need to pay for Claims.

This strategy doesn't work so well for shorter-term policies, such as ones for auto insurance, because insurance firms cannot invest in longer-term, illiquid Assets. As a result, companies that specialize in "Property & Casualty" (P&C) insurance look more like traditional companies.

But companies that offer longer-term policies with payouts decades into the future – such as life insurance firms – look more like commercial banks.



The Implications of Commercial Bank and Insurance Business Models

- 1) **Balance Sheet First** – The Balance Sheet drives these companies’ financial performance, and you often start the financial statements by projecting the Balance Sheet first. There are also some industry-specific items and different accounting conventions.
- 2) **Equity Value Only** – You cannot separate a bank or insurance firm’s **operating and financing activities** as cleanly as you can separate those of a traditional company. As a result, the concept of Enterprise Value (“The value of core-business operations, but to all investors”) does not apply to these companies. You use Equity Value and Equity Value-based multiples instead.
- 3) **Dividend Discount Models in Place of DCFs** – “Free Cash Flow” doesn’t mean anything for these firms because the Change in Working Capital and CapEx do not represent reinvestment in the business. So, you use Dividends as a proxy for FCF, Cost of Equity instead of WACC, and the Dividend Discount Model instead of the Discounted Cash Flow analysis.
- 4) **Regulations and Capital** – Banks and insurance firms are both highly regulated, and they must maintain minimum amounts of “capital” (Tangible Common Equity with a few modifications) at all times. These requirements constrain their growth.
- 5) **Different Valuation Multiples** – The Price / Book Value (P / BV), Price / Tangible Book Value (P / TBV), and Price / Earnings (P / E) multiples are all important because these firms are Balance Sheet-driven, and Interest is a huge part of their revenue. For life insurance firms, Price / Embedded Value (P / EV) is another important multiple.

The Fine Print

NONINTEREST INCOME

Trust and investment services income
Investment banking and debt placement fees
Service charges on deposit accounts
Operating lease income and other leasing gains
Corporate services income
Cards and payments income
Corporate-owned life insurance income
Consumer mortgage income
Mortgage servicing fees
Net gains (losses) from principal investing
Other income ^(a)

Total noninterest income

We’ve been assuming that banks **only** generate revenue via Net Interest Income.

But that’s not true: Look at a large bank’s Income Statement, and you’ll see revenue from “Service Charges,” “Corporate Services,” credit card fees, insurance, consumer mortgages, mortgage servicing fees, principal investing (trading), and a host of other categories.

These sources of revenue are so significant that banks create an entire category for them: **Non-Interest Income**.

The same is true of insurance firms, but an even higher percentage of their revenue comes from **Non-Interest/Investment sources** because of the **Premiums** they collect.



So, it's **not** as if banks and insurance firms are completely different from normal companies.

Instead, it's that since a *high* percentage of their revenue comes from Interest and Investment Income, we need to analyze, value, and model them differently.

This rule also means that if a "financial institution" does *not* earn a significant portion of its revenue from Interest, then it's pretty much the same as a traditional company.

That explains why wealth management firms, investment banks, brokerages, and fin-tech firms all follow traditional methodologies for accounting, financial statement projections, and valuation.

[Return to Top.](#)

Bank Accounting 101

Balance Sheet

It's still separated into Assets and Liabilities & Equity, but there are some important differences:

- **More Asset Classes:** Federal Funds Sold acts as a "balancer": It increases if the bank's L&E side has a surplus, and more Assets are needed for the BS to balance. Investments and Securities are often broken into many categories.
- **Gross Loans, Allowance for Loan Losses, and Net Loans:** "Gross Loans" is how much the bank has issued, total. The bank expects a certain number of borrowers to **default**, which is represented by the "Allowance for Loan Loss" contra-asset. Net Loans equals Gross Loans minus the Allowance for Loan Losses.
- **More Liabilities:** "Deposits" is the biggest one, but you also see Federal Funds Purchased (a "balancer" on the L&E side) and many types of Debt. Items like "Trading Liabilities" could represent **short positions** in securities (i.e., the bank is betting that a stock's price will decline).

Balance Sheet:	
ASSETS:	
Cash and Due from Banks:	
Investments and AFS/HTM Securities:	
Federal Funds Sold:	
Gross Loans:	
(-) Allowance for Loan Losses:	
Net Loans:	
Goodwill:	
Other Intangible Assets:	
Premises & Equipment:	
Other Assets:	
Total Assets:	
LIABILITIES & EQUITY:	
Liabilities:	
Interest-Bearing Deposits:	
Non-Interest-Bearing Deposits:	
Total Debt:	
Federal Funds Purchased:	
Other Liabilities:	
Total Liabilities:	
Equity:	
Common Shareholders' Equity:	
Preferred Stock:	
Noncontrolling Interests:	
Total Equity:	
Total Liabilities & Equity:	



Income Statement

A bank's Income Statement is split into three main sections: **Net Interest Income** (Interest Income minus Interest Expense), **Non-Interest Income**, and **Non-Interest Expenses**. There are also a few items in between these sections.

- **Non-Interest Income:** Investment banking fees, asset management fees, service charges, mortgage fees, credit card fees, commissions, and gains and losses on securities.
- **Non-Interest Expenses:** Employee compensation, rent, technology, marketing, professional services, amortization, and so on.

Income Statement:

Net Interest Income:

(+) Interest Income:

(-) Interest Expense:

Total Net Interest Income:

Non-Interest Revenue:

(+) Service Charges on Deposits:

(+) Trust Income:

(+) Other Fees and Leasing Income:

Total Non-Interest Revenue:

Total Net Revenue:

(-) Provision for Credit Losses:

Non-Interest Expenses:

(-) Personnel:

(-) Net Occupancy:

(-) Other Expenses:

(-) Amortization of Intangibles:

(-) Goodwill Impairment:

Total Non-Interest Expenses:

Pre-Tax Income:

(-) Income Tax Provision:

Net Income:

(-) Earnings from Discontinued Ops.:

(-) Noncontrolling Interest Earnings:

(-) Preferred Stock Dividends:

Net Income to Common:

Weighted Average Common Shares Outstanding:
Diluted EPS to Common:

- **Service Charges** are based on the bank's Deposits, but are *not* linked to prevailing interest rates: They're just a percentage of Total Deposits.
- **Trust Income** often refers to fees from wealth management and related services.
- **Provision for Credit Losses:** This number represents how much the bank **expects** to lose on its Loans in the current period **above the Allowance for Loan Losses** on its Balance Sheet. Banks do not record actual Loan losses on their Income Statements, only expected ones! (Although the two are very similar.)
- **Personnel:** Often the biggest Non-Interest Expense; salaries, benefits, training, and more.
- **Other Expenses:** Might include professional fees, marketing, and IT services.
- **Amortization of Intangibles and Goodwill Impairment:** Many banks are highly acquisitive, so these items are common.
- **Other Items:** Most large banks have Noncontrolling Interests and Preferred Stock, so you'll see items for those here; Discontinued Operations are also common since banks frequently restructure and reorganize.



Cash Flow Statement

Cash Flow Statement:

Operating Activities:

Net Income to Common:

Adjustments for Non-Cash Charges:

- (+) Provision for Credit Losses:
- (+) Depreciation & Amortization:
- (+) Amortization of Intangibles:
- (+) Goodwill Impairment:
- (+) Losses / (-) Gains:
- (+) Stock-Based Compensation:
- (+) Other Items:

Net Cash Flow from Operations:

Investing Activities:

- (+/-) Change in Investments and AFS/HTM Securities:
- (+/-) Change in Federal Funds Sold:
- (+/-) Change in Gross Loans:
- (+/-) Change in Other Assets:
- (-) Capital Expenditures (CapEx):

Net Cash Flow from Investing:

Financing Activities:

- (+/-) Change in Deposits:
- (+/-) Change in Total Debt:
- (+/-) Change in Federal Funds Purchased:
- (+/-) Change in Other Liabilities:
- (+/-) Change in Preferred Stock:
- (+/-) Change in Noncontrolling Interests:
- (-) Common Stock Dividends:
- (-) Excess Dividends / (+) Capital Infusion:
- (+) Common Stock Issuances:
- (-) Common Stock Repurchases:

Net Cash Flow from Financing:

Net Increase / (Decrease) in Cash:

Ending Cash on Balance Sheet:

Beginning Cash Balance:

Ending Cash Balance:

The main difference is that a bank's Cash Flow Statement could include items in different, seemingly random spots because there isn't a "strict" distinction between operating, investing, and financing activities.

The CFS still starts with Net Income (or, outside the U.S., often with Operating Income or Pre-Tax Income), and it still adjusts for major non-cash items, such as the Provision for Credit Losses, D&A, and Stock-Based Compensation.

It still includes familiar items, such as CapEx and Dividends, but their meaning differs: CapEx is relatively small for most banks and does **not** represent re-investment into the business. Dividends act as a proxy for the bank's "Free Cash Flow" but are also **constrained** by the bank's capital requirements.

The "Change in Deposits" and "Change in Gross Loans" here could easily appear in Operating Activities instead of Investing and Financing Activities.

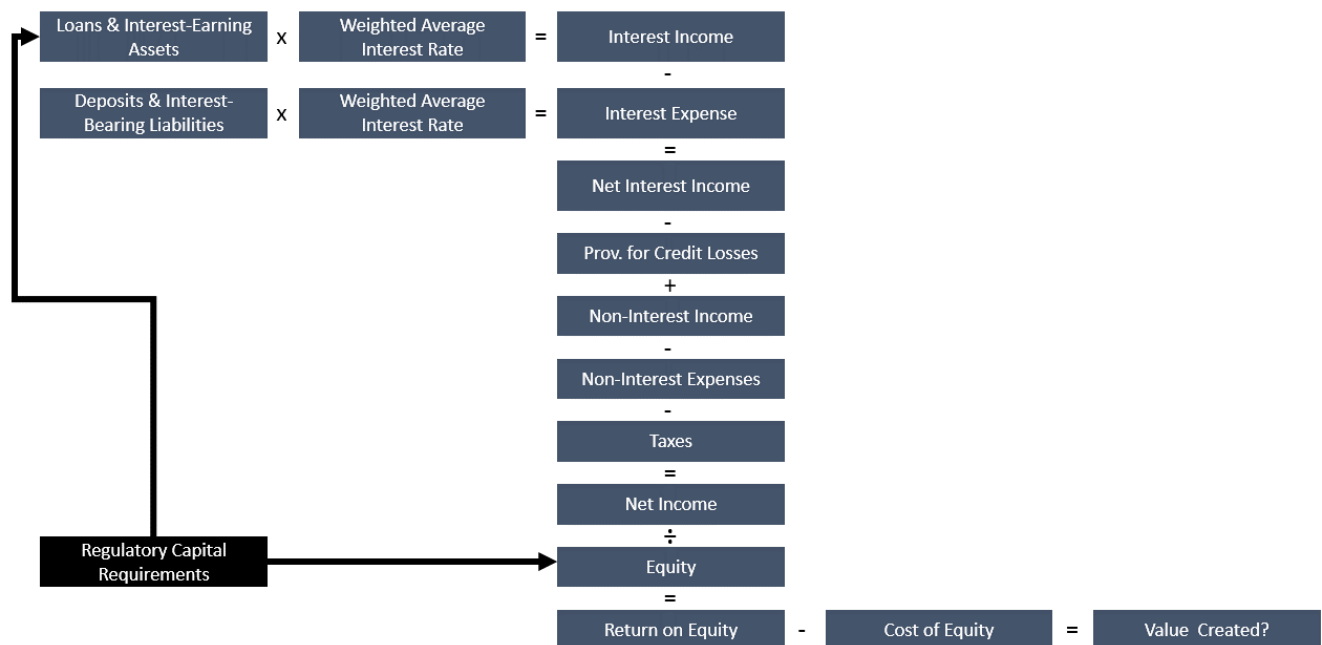
The Cash Flow Statement isn't terribly important for commercial banks because you *don't* need it for valuation purposes – just its Dividends, Net Income, and Book Value, all of which you can project without a Cash Flow Statement at all.

How Everything Links Together

- 1) You **always** start by projecting a bank's Balance Sheet, usually beginning with its Loans, Deposits, and other Interest-Earning Assets and Interest-Bearing Liabilities.
- 2) Then, project **interest rates** for all these items, and link them to a prevailing rate like LIBOR, the Federal Funds rate, or the interbank rate in your country.
- 3) Use that information to calculate the **Interest Income** and **Interest Expense** on the Income Statement. Estimate the Non-Interest Income and Expenses with simple percentage growth, percentage-of-revenue, or percentage-of-Balance-Sheet-line-item estimates.

- 4) Project the bank's **Dividends** based on the "capital" (roughly, Tangible Common Equity) it's targeting vs. how much it currently has. You can also complete the full Cash Flow Statement and include the other items as well, linking to the IS and BS where necessary.

Here's a diagram:



[Return to Top.](#)

Insurance Accounting 101



Although insurance firms and commercial banks are grouped together under the "Financial Institutions" moniker, insurance is a more complicated beast, with tricky accounting and new terminology to learn.

When you project insurance firms' financial statements, you start with the **Income Statement** because **Premiums** – not Loans or Deposits – drive all the other items.

The Balance Sheet is still important, and many insurance companies earn significant Interest and Investment Income from it, but **Premiums** act as the main driver.

Income Statement



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

Instead of using categories like COGS and Operating Expenses, here's how an insurance company might organize its Income Statement:

- **Revenue:** Premiums, Net Investment Income (mostly Interest), Net Investment Gains / (Losses), and Other Revenue.
- **Claims and Expenses:** Claim and Claim Adjustment Expenses (What they pay to customers who get in accidents), General & Administrative Expenses, Acquisition Costs (The commissions they're paying to acquire customers), and Interest Expense.
- **Pre-Tax Income** = Revenue – Claims and Expenses.
- **Net Income** = Pre-Tax Income * (1 – Tax Rate).

For the year ended December 31,

Revenues	
Premiums	
Net investment income	
Fee income	
Net realized investment gains (losses)	
Other revenues	
	Total revenues
Claims and expenses	
Claims and claim adjustment expenses	
Amortization of deferred acquisition costs	
General and administrative expenses	
Interest expense	
	Total claims and expenses
Income before income taxes	
Income tax expense	
	Net income

The trickiest part of an insurance company's Income Statement is **revenue and expense recognition**.

For example, let's say that a customer signs up for a \$10,000 1-year auto insurance policy on June 30th.

You'd record the \$10,000 as a **Written Premium**, but you would **not** recognize 100% of it as revenue in Year 1: Only half, or \$5,000, in **Earned Premiums** would count toward revenue for Year 1.

Insurance companies can recognize revenue only for policies they have *delivered*, so there is often a big difference between Written and Earned Premiums.

On the expense side, you might link **Claim and Claim Adjustment Expenses** to Earned Premiums – for example, maybe they represent 75% of Earned Premiums. In that case, you'd record \$3,750 in Claim and Claim Adjustment Expenses in this first year.

For 1-year policies, the math is straightforward. But the fun begins with policies that last for 2-3 years, 5-10 years, or even 20-30 years.

With multi-year policies, you still record on the Income Statement the **Claim and Claim Adjustment Expense** that matches up with the Earned Premiums in a given year.

But the *cash payout* of those expenses may be much different.



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

For example, if it's a 3-year policy with Premiums of \$1,000 per year, you might say that the Claim and Claim Adjustment Expense represents 75% of Earned Premiums each year.

So, you record \$750 in Year 1, \$750 in Year 2, and \$750 in Year 3, for a total expense of \$2,250.

But you might *pay out in cash* 60% of that \$2,250 figure in Year 1, 30% in Year 2, and 10% in Year 3, for Cash Expenses of \$1,350, \$675, and \$225.

On top of that, insurance companies often **reinsure** policies of *other* insurance companies, and, in turn, often hire *other* insurance companies to **reinsure** their policies (to distribute risk).

So, to calculate **Net Earned Premiums** – the figure that counts toward revenue – you must add Premiums the company is reinsuring ("Assumed Premiums") and subtract Premiums that *other* companies are reinsuring for this company ("Ceded Premiums").

Here's a screenshot of a partial model with some of this setup in place:

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Gross Written Premiums (GWP):		\$ 10,000	\$ 11,232	\$ 12,382	\$ 13,391	\$ 14,345
Gross Earned Premiums:		5,000	10,616	11,807	12,887	13,868
Ceded Written Premiums:		(3,000)	(3,370)	(3,715)	(4,017)	(4,303)
Ceded Earned Premiums:		(1,500)	(3,185)	(3,542)	(3,866)	(4,160)
Net Written Premiums (NWP):		7,000	7,862	8,668	9,374	10,041
Net Earned Premiums (NEP):		3,500	7,431	8,265	9,021	9,708
Change in Gross Unearned Premium Reserve:		5,000	616	575	505	477
Gross Unearned Premium Reserve:	-	5,000	5,616	6,191	6,696	7,172
Change in Ceded Unearned Premiums:		1,500	185	173	151	143
Ceded Unearned Premiums Balance:	-	1,500	1,685	1,857	2,009	2,152
Gross Losses & LAE Incurred:	-	3,750	7,885	8,686	9,480	10,202
Gross Losses & LAE Paid:		2,625	6,645	8,446	9,242	9,985
Gross Loss & LAE Reserve:	-	1,125	2,366	2,606	2,844	3,061
Ceded Losses & LAE Incurred:	-	1,125	2,366	2,606	2,844	3,061
Ceded Losses & LAE Paid:		788	1,993	2,534	2,773	2,996
Reinsurance Recoverables:	-	338	710	782	853	918
Net Losses & LAE Incurred:		2,625	5,520	6,080	6,636	7,141
Net Losses & LAE Paid:		1,838	4,651	5,912	6,469	6,990
Net Change in UEPR:		3,500	431	403	353	334

We use different types of **Reserves** to track revenue and expenses that have been recognized but not yet earned or paid in cash.

The **Unearned Premium Reserve (UEPR)** is the main one that tracks Premiums we've written but not yet earned; it's similar to Deferred Revenue for normal companies.

The **Loss & LAE Reserve** ("Loss & Loss Adjustment Expense" is another name for "Claim & Claim Adjustment Expense") does something similar for the company's main expense.

Finally, we must track the same items for the Premiums that have been **ceded** to other insurance firms; the Ceded Unearned Premiums and Reinsurance Recoverables line items on the Assets side do that.



Insurance Modeling = *Inception*?

Due to these complexities, insurance modeling sometimes feels like you're in the movie *Inception*: You think about policies within policies within policies... and then other companies insuring your policies, while other companies insure *their* policies.

However, most interview questions and case studies test your knowledge of the fundamentals, so don't get too paranoid about this complexity.

Balance Sheet

On the Balance Sheet, insurance companies differ from **both** commercial banks **and** normal companies.

The Balance Sheet is still divided into Assets and Liabilities & Equity, but the organization and individual items are very different:

Assets Side:

- **Investments:** Fixed Income, Equities, Real Estate, Joint Ventures, Short-Term Investments... anything on which the firm earns "Interest and Investment Income" goes here.
- **Non-Investment Assets:** Cash, Premiums Receivable (similar to Accounts Receivable), Reinsurance Recoverables, Ceded Unearned Premiums, Deferred Acquisition Costs, and then the usual items, like Goodwill and Other Intangible Assets.

Reinsurance Recoverables, Ceded Unearned Premiums, and Deferred Acquisition Costs are new items, but we need to look at the L&E side of the Balance Sheet first to explain what they mean:

At December 31,	
Assets	
Fixed maturities, available for sale, at fair value (amortized cost \$60,170 and \$63,311)	
Equity securities, available for sale, at fair value	
Real estate	
Short-term securities	
Other investments	
	Total investments
Cash	
Investment income accrued	
Premiums receivable	
Reinsurance recoverables	
Ceded unearned premiums	
Deferred acquisition costs	
Deferred tax asset	
Contractholder receivables	
Goodwill	
Other intangible assets	
Other assets	

Liabilities & Equity Side:

- **Liabilities:** Claim and Claim Adjustment Expense Reserves, Unearned Premium Reserves, Debt, Other Payables, and Other Liabilities.
- **Equity:** Preferred Stock, Common Stock, Noncontrolling Interests, APIC, Treasury Stock, Retained Earnings, and Accumulated Other Comprehensive Income. It's the same!

Liabilities

Claims and claim adjustment expense reserves
Unearned premium reserves
Contractholder payables
Payables for reinsurance premiums
Debt
Other liabilities

Shareholders' equity

Preferred Stock Savings Plan—convertible preferred
Common stock (1,748.6 shares authorized; 434.6 shares issued)
Retained earnings
Accumulated other changes in equity from non-owner transactions
Treasury stock, at cost (296.6 and 199.6 shares)

The two most important new line items are:

- **Claim and Claim Adjustment Expense Reserves (AKA “Loss and Loss Adjustment Expense Reserve”):** The Claim Expense on the Income Statement might differ significantly from the *cash* Claims paid to customers; this reserve reflects those differences. Each year, you **add** the Claim Expense on the Income Statement and **subtract** the Claims that are paid out in Cash. This reserve reflects **the total cash Claims the company must pay out in the future**.
- **Unearned Premium Reserves:** This item serves a similar function, but for Premiums rather than Claims. Each year, you **add** the Premiums the company has **written**, and you **subtract** the Premiums the company has **recognized** as revenue. This figure reflects the total amount of Premiums the company *will recognize* as revenue in the future.

Two of the new items on the **Assets side** are directly related to items on the L&E side:

- **Reinsurance Recoverables** refers to the Claim and Claim Adjustment Expense Reserve, but for policies the company has **ceded** to other insurance companies.
- **Ceded Unearned Premiums** refers to the Unearned Premium Reserve, but for policies the company has **ceded** to other insurance companies.

The last one, **Deferred Acquisition Costs**, is simpler: Each year, the company pays cash commissions to brokers and salespeople that referred new customers.

But the company can't recognize the entire cash expense right away since it does not recognize all the Written Premiums right away. Instead, it recognizes **only** the portion of the commission expense that corresponds to the *Earned Premiums* in the year.





The company defers the rest and recognizes it over time, and the Deferred Acquisition Costs Asset tracks how much will be recognized in the future (similar to Prepaid Expenses).

Cash Flow Statement

The Cash Flow Statement for insurance companies is similar to the CFS for normal companies: Start with Net Income (or sometimes other metrics, such as Operating Income or Pre-Tax Income), make non-cash adjustments, and factor in Changes in Working Capital to calculate Cash Flow from Operations.

Cash Flow from Investing is also similar: It includes purchases of and proceeds from investments and a bit of CapEx, often embedded in an “Other” line item.

Finally, Cash Flow from Financing also includes familiar items: Issuing and repaying Debt, issuing and repurchasing Stock, and issuing Dividends.

Since you also value insurance companies with a Dividend Discount Model rather than a DCF, the Cash Flow Statement is not terribly important for valuation purposes.

How Everything Links Together

Here’s a summary of how to project an insurance company’s financial statements:

1. Start by projecting the company’s **Direct Written Premiums** – how much in Premiums it is selling directly to customers. Then, project the company’s **Assumed Premiums** and **Ceded Premiums**, and the percentage that it’s earning in each category.
2. Next, project the **Claim and Claim Adjustment Expense Ratio**, the commission rate, and the underwriting expenses as a percentage of the Written.
3. List the company’s **Net Earned Premiums** and its Interest and Investment Income for revenue on the Income Statement. You’ll need the Balance Sheet to get the full numbers for Interest and Investment Income. Expenses consist of Claims, Commissions, Underwriting Expenses, and Interest. Calculate Pre-Tax Income and Net Income the normal way.
4. The **Balance Sheet** flows in from the company’s revenue/expense recognition (the Reserves and corresponding items on the Assets side), and Cash and Equity flow in as they normally do. You might project Investments and Debt independently, or you might link them to Premiums.

5. The **Cash Flow Statement** works the same way as well: Start with Net Income, make non-cash adjustments, factor in Changes in Working Capital, and sum up each section of the CFS to calculate the Net Change in Cash at the bottom.
6. Return to the **Income Statement** and link in Interest/Investment Income and Interest Expense, which you projected based on Balance Sheet numbers and separate assumptions.

[Return to Top.](#)

Regulatory Capital



Both banks and insurance firms invest *customers'* money to make money – which means that they could also take on too much risk and **lose** their customers' money.

As a result, **regulations** in both industries limit the risk-taking potential of these firms.

Of course, both types of companies **expect** to lose some amount of money over time. Banks reflect this expectation with the **Allowance for Loan Losses**; insurance companies do it with their **Claims Reserve**.

“Regulatory Capital” exists to cover unexpected losses – it’s a “cushion” that ensures that even if something goes horribly wrong, the firm will survive (in theory).

Here are the most important Regulatory Capital metrics you need to know:

- **Banks:** Common Equity Tier 1, Tier 1 Capital, Total Capital, Liquidity Coverage Ratio, and Net Stable Funding Ratio.
- **Insurance:** Solvency Capital Requirement Coverage Ratio.

Common Equity Tier 1 (CET 1) for banks is Tangible Common Equity (Common Shareholders' Equity minus Goodwill minus Other Intangible Assets), plus or minus a few other adjustments.

It serves as a buffer against potential *unexpected* losses.

Think about what happens if a bank writes down a Loan that it did *not* expect to write down: If something on the Assets side decreases, something on the Liabilities & Equity side must also decrease.



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

If the bank does **not** have a capital buffer, that “something” might be Deposits or Debt – both of which would be bad (customers and lenders would stop trusting the bank).

So, Common Equity Tier 1 and its variants ensure that **you, the consumer, do not lose money in your checking account when a bank experiences unexpected Loan losses.**

Tier 1 Capital equals CET 1 + Preferred Stock. And Total Capital equals Tier 1 Capital + “Tier 2 Capital.”

Tier 2 Capital includes some hybrid instruments (e.g., Convertible Bonds), Subordinated Notes, and a portion of the Allowance for Loan Losses.

It’s known as “Gone Concern Capital” because the items within it can absorb losses only if the bank stops operating (i.e., there’s a bankruptcy or other shutdown).

You always divide these metrics by some variant of Total Assets: The two most common ones are **Risk-Weighted Assets**, which “weights” the Assets to reflect the fact that Loans to the government of Greece are far riskier than Loans to Google, and **Tangible Assets**.

For example, the “CET 1 Ratio” equals Common Equity Tier 1 / Risk-Weighted Assets. The “Leverage Ratio” equals Tier 1 Capital / Tangible Assets (+/- various adjustments).

Banks must maintain capital ratios above certain levels at all times to avoid penalty fees, restrictions on Dividends and Stock Repurchases, and other limitations. If their capital ratios fall to too low a level, the government may come in and take over.

For example, under Basel III, the minimum CET 1 Ratio is 4.5%; with the “capital conservation buffer,” that climbs to 7.0%, and with the “countercyclical buffer” included, it’s 9.5%.

The minimum Tier 1 Capital Ratio is 6.0%, and the minimum Total Capital Ratio is 8.0% (both would also increase by 2.5% or 5.0% with the buffer(s) included).

Example - Regulatory Capital Calculations for JP Morgan (\$ in Millions)

JP Morgan - Regulatory Capital and RWA Calculations

Total Stockholders' Equity:	\$ 232,065
(-) Preferred Stock:	(20,063)
Common Stockholders' Equity:	212,002
(-) Goodwill:	(44,925)
(-) Other Intangible Assets:	(1,062)
(-) Other CET1 Capital Adjustments:	(1,163)
Common Equity Tier 1 (CET 1) Capital:	164,852
(+) Preferred Stock:	20,063
(-) Other Tier 1 Adjustments:	(5)
Total Tier 1 Capital:	184,910
(+) LT Debt & Others Qualifying as Tier 2 Capital:	17,504
(+) Qualifying Allowance for Credit Losses:	4,266
(-) Other:	(86)
Total Tier 2 Capital:	21,684
Total Capital:	\$ 206,594
Credit Risk RWA:	\$ 1,040,087
Market Risk RWA:	179,200
Operational Risk RWA:	400,000
Total Risk-Weighted Assets:	\$ 1,619,287
CET 1 Ratio:	10.2%
Tier 1 Capital Ratio:	11.4%
Total Capital Ratio:	12.8%



Banks also have to comply with requirements for **liquidity** and **stability**.

The **Liquidity Coverage Ratio (LCR)** requires banks to maintain enough “high-quality assets” to cover 100% of net cash outflows during a 30-day “stressed period.” To measure that, you examine the bank’s liquid Assets (primarily Cash) and compare them to the percentage of Deposits and other funding sources that might dry up in a crisis.

The **Net Stable Funding Ratio (NSFR)** requires banks to maintain “available stable funding” that equals or exceeds the “required stable funding” over a 1-year period.

“Available Stable Funding” is, roughly, Total Capital plus various Liabilities and funding sources with maturities of 1 year or more; “Required Stable Funding” consists of the bank’s Assets plus off-Balance Sheet commitments, weighted by various percentages.

Insurance is similar, but a bit simpler: The key ratio is the “Solvency Capital Requirement Coverage Ratio,” or SCR Coverage Ratio, which is defined as Available Capital / Required Capital.

Insurance companies must keep that ratio above 100% at all times, but most firms stay comfortably above it, with median percentages frequently in the 200-300% range.

“Available Capital” is similar to Total Capital for commercial banks: It’s mostly Tangible Common Equity along with some longer-term Liabilities and funding sources.

Required Capital equals “the expected negative impact on Own Funds of a 1-in-200-year event,” so you cannot calculate it directly with a simple formula. However, it is related to the firm’s Total Assets and the amount of risk it has taken on via cumulative Written Premiums.

There is also a “Minimum Capital Requirement” (MCR) for insurance firms, which is similar to the SCR Coverage Ratio but set to a 25-45% minimum rather than a 100% minimum.

This one is like “the point of no return” for insurance firms; if a firm approaches this level, regulators will step in to run the business.

Why Does Regulatory Capital Matter?



It matters because it **constrains the assumptions in operating models and valuations for financial institutions.**

You can't blindly assume a Loan or Premium Growth Rate – you must ensure that the firm meets its regulatory capital requirements no matter what you do.

So, for example, if you assume that a bank grows its Loans at 30% per year, you might find that the bank's CET 1 Ratio falls below its 10% targeted level by Year 5.

If that happens, you need to assume slower Loan Growth or assume that the bank raises additional equity via a follow-on offering so that its CET 1 Ratio complies with the minimum.

These requirements also influence the Dividends that banks and insurance firms can issue: Dividends always reduce Equity, and all regulatory capital ratios are linked to Equity.

So, in the scenario above, another solution might be to keep the 30% Loan Growth Rate, but assume that the bank reduces its targeted Dividend Payout Ratio.

Since you value both banks and insurance firms with the **Dividend Discount Model**, these requirements directly influence the implied values from that methodology.

You can't just say, "This bank will grow its Loans at 30% per year and issue Dividends representing 50% of its Net Income each year." That assumption **only** works if the bank also complies with its regulatory capital requirements.

Banks often raise Debt or Equity, divest divisions, or acquire other companies because of regulatory capital concerns. You might even adjust the comparable public companies for excess or deficit capital or screen for companies or transactions based on regulatory capital.

[Return to Top.](#)

Valuation Multiples & Intrinsic Valuation



The two key valuation multiples for both banks and insurance firms are **P / E** (Price per Share / Earnings per Share, or Equity Value / Net Income) and **P / BV** (Price per Share / Book Value per Share, or Equity Value / Book Value).

A third common multiple is **P / TBV** (Price per Share / Tangible Book Value per Share, or Equity Value / Tangible Book Value); Tangible Book Value subtracts Goodwill and Other Intangible Assets from the normal Book Value metric.

P / E measures the firm's value relative to its after-tax profitability. You use Net Income rather than EBITDA or EBIT because you **want** to include Net Interest Income for these companies. Also, Net Income and Dividends tend to be closely related for financial institutions.



P / BV and P / TBV are important because they are strongly correlated with a firm's ROE (Return on Equity) and ROTCE (Return on Tangible Common Equity), respectively.

In other words, if a firm's ROE is higher than the median ROE of its peer companies, then its P / BV multiple *should* be higher than the median P / BV of the set. If it is not, then the firm might be undervalued.

You can even link these metrics formulaically and say that, for a stabilized firm:

$$P / BV = (\text{Return on Equity} - \text{Net Income Growth Rate}) / (\text{Cost of Equity} - \text{Net Income Growth Rate})$$

If the P / BV multiple is above 1x, ROE exceeds the Cost of Equity; if it's below 1x, ROE is below the Cost of Equity (in theory...).

As with normal companies, you also select and use **comparable public companies** and **precedent transactions** for banks and insurance firms.

But the selection criteria are different: Instead of screening by Revenue or EBITDA, you might screen based on Loans, Deposits, Premiums, Total Assets, or other Balance Sheet items.

Also, the **geography** is extremely important because both industries are highly regulated, and the rules may differ significantly in different countries.

Other Variations of Multiples



You will see non-recurring adjustments for P / E multiples, as well as adjustments for "excess or deficit capital," which apply to all the multiples above.

In other words, if a bank has too much CET 1 (e.g., a 15% ratio vs. 10% for its peers), you might subtract the excess CET 1 from the bank's Equity Value, Book Value, and Tangible Book Value, reduce its Net Income based on the earnings from that excess capital, and then recalculate all the multiples.

In insurance, if the firm does **not** mark its Balance Sheet to market value, you might see P / NAV (“Net Asset Value”) multiples as well.

The “Net Asset Value” is an estimate of the market value of the firm’s Assets minus the market value of its Liabilities.

Embedded Value and P / EV multiples are also important for life insurance companies, but we’ll cover those below.

Intrinsic Valuation

The most important intrinsic valuation methodology for banks and many insurance firms is the **Dividend Discount Model**.

“Free Cash Flow” is meaningless for these companies because CapEx and the Change in Working Capital do **not** represent re-investment in their core businesses – so you cannot use a traditional DCF to value them.

Instead, you use **Dividends** as a proxy for Free Cash Flow. Since Dividends are only available to common equity investors, you use Cost of Equity for the Discount Rate, and you base the Terminal Value on multiples such as P / E, P / BV, or P / TBV.

Here’s how you might set up a simple Dividend Discount Model (DDM) for a commercial bank:

1. Assume an **Asset Growth Rate**, and make **Risk-Weighted Assets** a percentage of Total Assets.
2. **Project** Assets and Risk-Weighted Assets based on these figures.
3. Assume a **Return on Assets (ROA)** for the bank, and use that to project Net Income. $ROA = \text{Net Income} / \text{Average Total Assets}$, so $\text{Net Income} = ROA * \text{Average Total Assets}$.
4. Then, assume a **Targeted CET 1 Ratio** (e.g., 10% or 12%), and calculate the bank’s CET 1 for the year based on that percentage times its Risk-Weighted Assets.
5. **Back into the Dividends Issued** by taking the bank’s CET 1, adding Goodwill and Other Intangibles, subtracting the beginning Shareholders’ Equity, and subtracting Net Income and anything else that might affect CET 1.





Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

You **constrain** the Dividends issued by setting it up this way.

6. **Discount** the Dividends based on the Cost of Equity, and sum up all the discounted values.
7. Calculate the **Terminal Value** using the Multiples Method (typically P / TBV) or the Gordon Growth method, and discount it to its Present Value, also using the Cost of Equity.
8. **Add** the Present Value of the Dividends and the Present Value of the Terminal Value to determine the bank's Implied Equity Value. Compare this to the bank's Current Equity Value.

Step 5 is the crucial difference between a DDM and a DCF: You can't make arbitrary assumptions about growth, profitability, or payout ratios. If you do, and the CET 1 falls below the target, the bank will not even be able to issue Dividends.

Here's what a simplified DDM might look like:



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

Simplified DDM Assumptions:

Targeted CET 1 Ratio:	8.0%
Total Asset Growth:	4.0%
RWA % Total Assets:	75.0%
Return on Assets (NI to Common):	0.55%
Cost of Equity:	9.0%
Stock Issuances:	-
Stock Repurchases:	-

Terminal Value Calculation:

Baseline Terminal P / TBV Multiple:	1.03 x
Baseline Terminal Value:	153
(+) PV of Terminal Value:	84
(+) PV of Dividends:	35
Implied Equity Value:	\$ 119
Bank's Current Equity Value:	\$ 150

Premium / (Discount) to Implied Value	26.3%
--	--------------

Dividend Projections:	Historical			Projected						
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Net Income to Common:	\$ 9	\$ 9	\$ 8	\$ 10	\$ 10	\$ 11	\$ 12	\$ 12	\$ 13	\$ 13
% Growth:		(3.1%)	(9.3%)	22.8%	4.6%	6.3%	5.8%	4.0%	4.0%	4.0%
Dividends:	5	5	5	6	7	7	8	7	7	7
% Growth:		0.0%	0.0%	24.1%	6.3%	7.9%	14.9%	(16.9%)	4.0%	4.0%
Payout Ratio:	54.8%	56.5%	62.4%	63.0%	64.0%	65.0%	70.5%	56.4%	56.4%	56.4%
Beginning Common Shareholders' Equity:	N/A	160	166	171	178	184	191	194	200	205
(+) Net Income to Common:	9	9	8	10	10	11	12	12	13	13
(-) Common Stock Dividends:	(5)	(5)	(5)	(6)	(7)	(7)	(8)	(7)	(7)	(7)
(+) Stock Issuances:	3	4	3	4	5	5	-	-	-	-
(-) Stock Repurchases:	(1)	(2)	(1)	(1)	(2)	(2)	-	-	-	-
Ending Common Shareholders' Equity:	160	166	171	178	184	191	194	200	205	211
(-) Goodwill:	(50)	(49)	(48)	(48)	(48)	(48)	(48)	(48)	(48)	(48)
(-) Other Intangible Assets:	(20)	(19)	(18)	(17)	(16)	(15)	(15)	(15)	(15)	(15)
Common Equity Tier 1 (CET 1):	90	98	105	113	120	128	131	137	142	148
Total Assets:	1,685	1,774	1,850	1,936	2,023	2,107	2,191	2,279	2,370	2,465
Growth Rate:		5.3%	4.2%	4.6%	4.5%	4.1%	4.0%	4.0%	4.0%	4.0%
Risk-Weighted Assets:	1,200	1,300	1,375	1,438	1,516	1,593	1,643	1,709	1,777	1,848
RWA % Total Assets:	71.2%	73.3%	74.3%	74.3%	75.0%	75.6%	75.0%	75.0%	75.0%	75.0%
CET 1 Ratio:	7.5%	7.5%	7.6%	7.8%	7.9%	8.0%	8.0%	8.0%	8.0%	8.0%
Return on Tangible Common Equity:	N/A	9.8%	8.2%	9.4%	9.2%	9.1%	9.0%	9.2%	9.2%	9.2%
Return on Assets (NI to Common):	N/A	0.52%	0.45%	0.53%	0.53%	0.54%	0.55%	0.55%	0.55%	0.55%
Discount Period:			0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0
PV of Dividends:				\$ 6	\$ 6	\$ 5	\$ 6	\$ 4	\$ 4	\$ 4

In real life, the Targeted CET 1 Ratio would probably be higher; the Total Asset Growth might also drop to less than 4% by the end of the projection period.

[Return to Top.](#)

Bank-Specific and Insurance-Specific Valuation

Valuation – Bank-Specific

You use the DDM outlined above 99% of the time when valuing banks.



But there is another, closely related methodology, known as the **Residual Income** or **Excess Returns Model**.

With that one, the starting point is the bank's **Book Value**: You assume that the bank's Implied Equity Value is initially equal to its Book Value.

Then, if the bank earns *excess returns* – if its Return on Equity (ROE) exceeds its Cost of Equity (K_e) in future periods – you calculate, discount, and add all these “excess returns” to the Book Value.

If the bank earns some amount of excess returns, its *implied P / BV* should be above 1x; if not, it should be less than or equal to 1x.

You set up this model almost the same way you set up a Dividend Discount Model – you still assume a Targeted CET 1 Ratio, and you still back into Dividends based on the bank's actual vs. targeted CET 1.

But you do **NOT** value the bank based on those Dividends.

Instead, you calculate the **Residual Income** or **Excess Returns** each year with: $(ROE * \text{Shareholders' Equity} - K_e * \text{Shareholders' Equity})$. And then you discount and sum up those values and add them to the bank's current Book Value to determine its Implied Equity Value.

With this methodology, most of the bank's Implied Value comes from its *current* Balance Sheet, so it's less speculative than a DDM. It also works well for banks that are not currently issuing Dividends.

But the disadvantage is that it's far less common than the DDM, so it will require some explanation if you decide to use it. Many bankers also think of this methodology as “academic” and not as useful in real life.

Valuation – Insurance-Specific

Embedded Value is an important valuation methodology for **life insurance** companies.

It's similar to the Residual Income Model described above because you take the insurance firm's Balance Sheet as the starting point in the valuation, and then you add the present value of **future cash profits** from the insurance firm's *current policies*.

Since life insurance contracts last for much longer than property & casualty (P&C) contracts, you can reasonably project cash flows and profits 20-30 years into the future.



To calculate Embedded Value, start with the firm's **Net Asset Value** and add the present value of all expected, *future* cash profits.

"Net Asset Value" means that you mark the insurance firm's Balance Sheet to market value if it is **not** yet marked-to-market, and then you subtract all the Liabilities from all the Assets.

You can also think of Net Asset Value as the cumulative, after-tax cash flows generated from policies in previous years.

So, Embedded Value = Cumulative, After-Tax Cash Flows from Policies in Past Years + Present Value of Expected After-Tax Cash Flows in Future Years.

When you calculate Embedded Value *on today's date*, you assume that the insurance company writes **no new policies** in future years. The expected after-tax cash flows, therefore, are all based on the company's *policies as of today*.

In real life, you may *project* Embedded Value by assuming that the company writes new policies in each future year.

HOWEVER, projected Embedded Value in Year 4 or Year 10 is still based on the assumption that the company writes no new policies after Year 4 or Year 10.

Embedded Value is an intrinsic valuation methodology, but you can also calculate a **P / EV** (Equity Value / Embedded Value) multiple based on this analysis.

It's one of the most important multiples for life insurance companies, and you often use it in addition to or in place of P / E and P / BV.

[Return to Top.](#)

M&A and Merger Models



You shouldn't worry about these topics too much because they are **extremely** unlikely to come up in interviews.

But since you asked, here are the main differences in **bank M&A deals and merger models** (sorry, I've never been able to find much on insurance):

- 1) **Form of Payment** – Most banks can use only **Debt** or **Stock** to fund deals; they need to keep significant Cash available to cover possible Deposit withdrawals. Sometimes other banks also hold a significant percentage of the bank's Cash.
- 2) **Mark-to-Market Balance Sheet** – In any M&A deal, the Buyer must mark the Seller's Balance Sheet to market value, which usually means adjusting the value of the Seller's Loans ("Loan Marks") because of interest rate or credit risk differences. The combined company must then amortize these adjustments over the estimated useful lives of the Assets and Liabilities:

Mark-to-Market Schedule for Seller's Balance Sheet - Target Thrift

(\$ in Millions USD Except Per Share Data)

Assumptions:

Balance Sheet Item:	Market Value:	Adj. Value at Close:	Increase / (Decrease):	Amort. Period:	Amort. Amount:
Investments and AFS/HTM Securities:	99.0%	188.1	(1.9)	5.0	0.4
Federal Funds Sold:	100.0%	10.0	-	5.0	-
Gross Loans:	99.0%	356.4	(3.6)	5.0	0.7
Deposits:	99.0%	341.6	(3.4)	2.0	(1.7)
Total Debt:	99.0%	99.0	(1.0)	5.0	(0.2)
Federal Funds Purchased:	100.0%	55.0	-	5.0	-
Total:	99.1%	1,050.1	(10.0)		(0.8)

Amortization of Mark-to-Market Adjustments:	Year 4	Year 5	Year 6	Year 7	Year 8
Investments and AFS/HTM Securities:	0.4	0.4	0.4	0.4	0.4
Federal Funds Sold:	-	-	-	-	-
Gross Loans:	0.7	0.7	0.7	0.7	0.7
Deposits:	(1.7)	(1.7)	-	-	-
Total Debt:	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)
Federal Funds Purchased:	-	-	-	-	-
Total:	(0.8)	(0.8)	0.9	0.9	0.9

- 3) **Write-Down of the Allowance for Loan Losses** – Accounting rules require the Buyer to write down the Seller's Allowance for Loan Losses since it is also marking the Loan portfolio to fair market value.



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

- 4) **Core Deposit Intangibles** – A new Intangible Asset, which represents the “funding advantage” of the Seller’s most stable Deposits, gets created. The combined company amortizes it over time, and the amortization is typically not deductible for cash-tax purposes.

Sources & Uses of Funds:

Sources:	
Senior Debt Issued:	\$ 13.5
Subordinated Debt Issued:	4.5
Common Stock Issued:	161.5
Cash for Transaction Fees:	3.6
Total Sources:	\$ 183.0

Uses:	
Equity Purchase Price of Target:	\$ 179.4
Transaction Fees:	3.6
Total Uses:	\$ 183.0
CHECK:	OK!

Purchase Price Allocation:

Goodwill Calculation:	
Equity Purchase Price:	\$ 179.4
(-) Seller Common Book Value:	(113.5)
(+) Write-Off of Existing Goodwill:	20.0
Total Allocable Purchase Premium:	85.9
(+/-) Mark-to-Market Adjustments on Seller's BS:	1.1
(-) Write-Down of Seller's Allowance for Loan Losses:	(6.5)
(+) Write-Down of Seller's Intangibles:	21.0
(-) Core Deposit Intangibles (CDI) Created:	(12.1)
(-) Write-Up of Intangibles:	(4.3)
(+) New Deferred Tax Liability:	5.7
Total Goodwill Created:	\$ 90.8

Core Deposit Intangibles:	
Core Deposits % Total:	70.0%
Total Core Deposits:	241.5
Core Deposit Intangibles % Core Deposits:	5.0%
Core Deposit Intangibles:	12.1
Amortization Period (Years):	10.0

Intangible Asset Write-Up:	
Purchase Price to Allocate:	\$ 85.9
% Allocated to Indefinite-Lived Intangibles:	5.0%
Indefinite-Lived Intangibles:	4.3
New Deferred Tax Liability:	\$ 5.7

- 5) **Deposit Divestitures** – The Buyer may have to divest some of the Seller’s Loans and Deposits for regulatory reasons. For example, in the U.S., banks cannot end up owning over 10% of total national deposits as a result of M&A deals (but they can reach that level organically).
- 6) **Cost Savings and Restructuring** – Cost savings opportunities are a bit easier to predict in bank M&A deals because most commercial banks have the same business model. In many deals, Buyers plan to cut 20%, 30%, or even 40% of the Seller’s Non-Interest Expenses, which would be massive in other industries. Buyers must spend some upfront cash on Restructuring to realize these Cost Savings.
- 7) **Capital Ratios, Dividends, and Federal Funds** – The Buyer’s Regulatory Capital, Dividends, and Federal Funds Sold and Purchased will all be impacted by the deal. Depending on the numbers, the Buyer may have to raise more Equity, increase or reduce Dividends, and increase or reduce its Federal Funds line items.
- 8) **Alternative Assessments** – Beyond EPS accretion/dilution, you can also look at the accretion/dilution of metrics such as Dividends per Share and Book Value per Share, and you



can evaluate deals by calculating the Internal Rate of Return (IRR) or the Net Present Value (NPV) of Synergies. A Contribution Analysis is also useful for majority-Stock deals.

A merger model is still a merger model, so you still combine all the financial statements, allocate the purchase price, and so on – it's just that the new items above complicate the model.

For example, the Purchase Price Allocation Schedule becomes more involved because you also have to factor in the mark-to-market adjustments, the write-down of the Allowance for Loan Losses, and the new Core Deposit Intangibles when calculating Goodwill.

[Return to Top.](#)

Buyout and Growth Equity Models

Traditional “leveraged buyouts” are quite rare for commercial banks (they're more common in other sectors of financial services, including insurance).

That's because of **regulations** and **deal math**:

- 1) **Regulations:** If a private equity firm comes to own more than 25% (or sometimes even less) of a bank's voting shares, the PE firm may be classified as a “bank holding company,” which means that it must comply with regulatory capital and other requirements.
- 2) **Deal Math:** Most banks are already highly leveraged and cannot take on much additional Debt to fund a deal. Also, if too much Debt is used, the bank's CET 1 Ratio could fall below the minimum (since the bank's Equity is written down 100% in a control deal).

To get around these problems, PE firms typically invest in the sector in one of three ways:

- **Option #1:** Structure the deal as a minority-stake investment for 5%, 10%, or 20% of the bank's common shares.
- **Option #2:** Invest in something *other* than the bank's Common Equity – Preferred Stock, Convertibles, or Mezzanine. Or purchase specific Assets, such as a portion of the bank's Loan portfolio.
- **Option #3:** Do the deal with multiple PE firms such that it *is* a buyout for 100% of the bank, but *no firm* controls more than, say, 20% of the bank's voting shares. The deal would be funded with 100% Equity, or a percentage close to 100%.

“Bank buyouts” are more like **growth equity deals** or **debt investments** than traditional LBOs.

The differences are clear if you think about the **sources of returns** in a traditional LBO model vs. a “bank buyout” model:

Traditional LBO Model – Returns Attribution Analysis:

Value Creation Analysis:		Projected				
		FY16	FY17	FY18	FY19	FY20
Value Creation in EUR:						
EBITDA Growth:	EUR m	35.0	367.4	684.5	958.8	1,207.0
Multiple Expansion:	EUR m	-	211.2	454.1	722.2	1,012.6
Debt Paydown and Cash Generation:	EUR m	23.1	117.5	272.4	438.0	634.9
Total Equity Return:	EUR m	58.1	696.0	1,411.0	2,119.0	2,854.5
Value Creation in %:						
EBITDA Growth:	%	60.3%	52.8%	48.5%	45.2%	42.3%
Multiple Expansion:	%	0.0%	30.3%	32.2%	34.1%	35.5%
Debt Paydown and Cash Generation:	%	39.7%	16.9%	19.3%	20.7%	22.2%
Total Equity Return:	%	100.0%	100.0%	100.0%	100.0%	100.0%

This Returns source goes away in "bank buyouts."

“Bank Buyout” Model – Returns Attribution Analysis:

Returns Attribution Analysis:	Amount:	%:
TBV Growth:	\$ M 1,910.4	46.7%
P / TBV Multiple Expansion:	\$ M 1,008.5	24.6%
Dividends:	\$ M 1,175.8	28.7%
Total Return to Equity Investors:	\$ M 4,094.7	100.0%

Same idea as with traditional companies and LBOs, but different metrics and multiples.

Dividends aren't really "new" - they could still be a source of returns in traditional LBOs. But they're more common and important in bank buyouts/investments.

The goal in almost any bank buyout or minority investment is to boost the bank’s **P / TBV** or **P / BV** multiple. Assuming a stabilized bank, you can calculate these multiple with formulas:

$$P / BV = (ROE - NI \text{ Growth Rate}) / (\text{Cost of Equity} - NI \text{ Growth Rate})$$

$$P / TBV = (ROTCE - NI \text{ to Common Growth Rate}) / (\text{Cost of Equity} - NI \text{ to Common Growth Rate})$$

Most PE firms attempt to boost banks’ ROTCE or ROE via cost cuts, higher Asset growth, higher-yielding Assets, or lower-cost funding sources.

A simplified outline of a “bank buyout” deal might look like this:



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

Income Statement:	Units:	Historical			Transaction Adjustments			Projected				
		FY 14	FY 15	FY 16	Debit	Credit	FY 16	FY 17	FY 18	FY 19	FY 20	FY 21
Pre-Transaction Return on Assets (ROA):	%	2.3%	2.8%	2.7%				2.4%	2.2%	2.1%	2.1%	2.1%
(+) ROA Improvement:	%							0.2%	0.2%	0.2%	0.2%	0.2%
Post-Transaction Return on Assets (ROA):	%							2.6%	2.4%	2.3%	2.3%	2.3%
Net Income to Common:	\$ M	\$ 225.8	\$ 272.7	\$ 315.4				\$ 392.0	\$ 413.3	\$ 438.9	\$ 477.3	\$ 519.4
Net Income to Common Growth Rate:	%	(17.7%)	20.8%	15.7%				24.3%	5.4%	6.2%	8.8%	8.8%
Common Dividends Issued:	\$ M							-	173.4	297.8	328.3	362.6
Dividend Payout Ratio:	%							0.0%	42.0%	67.8%	68.8%	69.8%
Return on Tangible Common Equity (ROTCE):	%		28.7%	27.9%				36.8%	29.5%	27.3%	26.9%	26.7%

Investor Returns:	Units:	Historical			Transaction Adjustments			Projected				
		FY 14	FY 15	FY 16	Debit	Credit	FY 16	FY 17	FY 18	FY 19	FY 20	FY 21
Net Cash Flows to Investors:												
(+/-) Investor Equity:	\$ M					\$ (3,132.0)		\$ -	\$ -	\$ -	\$ -	\$ 6,109.4
(+) Dividends:	\$ M					-		-	173.4	297.8	328.3	362.6
Total Net Cash Flows:	\$ M					\$ (3,132.0)		\$ -	\$ 173.4	\$ 297.8	\$ 328.3	\$ 6,472.0
Money-on-Money (MoM) Multiple:	x					2.3 x						
Internal Rate of Return (IRR):	%					19.3%						

Returns Attribution Analysis:		Amount:	%:
TBV Growth:	\$ M	\$ 1,959.2	47.3%
P / TBV Multiple Expansion:	\$ M	1,018.2	24.6%
Dividends:	\$ M	1,162.0	28.1%
Total Return to Equity Investors:	\$ M	\$ 4,139.5	100.0%

[Return to Top.](#)

Recap and Summary

Here's a summary of the key points in this guide:

- **How Banks & Insurance Firms Work:** They make money *with* money; they accept Deposits or Premiums from customers, pay Interest or Claims on them (eventually), and then they invest the customers' funds in Loans or Investments and earn money on those; they also earn money from commissions, fees, and other non-Interest sources.
- **Bank Accounting 101:** The key items on a bank's Balance Sheet are Loans and Deposits; Gross Loans minus the Allowance for Loan Losses (what the bank *expects* to lose) equals Net Loans; banks have many investment/asset categories and funding sources; the Income Statement is split into Non-Interest vs. Interest Revenue and Expenses, and the Provision for Credit Losses represents what the bank expects to lose *over and above* the Allowance on the BS.
- **Insurance Accounting 101:** The Income Statement is split into Revenue (Premiums and Investment/Interest Income) and Claims & Expenses; revenue/expense recognition is tricky because policies can last many years but are often paid for upfront; the Balance Sheet has Investment/Non-Investment Assets, and the L&E side has Reserves for Claims and Claim



Bank & Financial Institution Modeling

Quick Reference Guide for the Entire Course

Adjustment Expenses and Unearned Premiums; companies often assume premiums from and cede premiums to other insurance companies.

- **Regulatory Capital:** Common Equity Tier 1 (Tangible Common Equity with a few adjustments), Total Capital, etc. all protect commercial banks from *unexpected* losses; if a bank suddenly has to write down its Loans, the losses will affect Shareholders' Equity rather than Deposits or Debt. For insurance, the Solvency Capital Requirement Coverage Ratio, defined as Available Capital / Required Capital, is similar.
- **Valuation Multiples & Intrinsic Valuation:** P / E , P / BV , and P / TBV are key multiples for both commercial banks and insurance firms; the Dividend Discount Model is used in place of a DCF, and all assumptions must be tied to regulatory capital, i.e. Loans and Deposits can grow only if the bank has enough Equity to support the additional Risk-Weighted Assets. You *back into* the Dividends issued each year based on Targeted CET 1 vs. Actual CET 1.
- **Bank-Specific and Insurance-Specific Valuation:** For banks, you may use a Residual Income Model, which uses Book Value + Present Value of Future Residual Income (Residual Income = $ROE * \text{Shareholders' Equity} - \text{Cost of Equity} * \text{Shareholders' Equity}$ in each year) to value the company. For life insurance, you may use Embedded Value, which values a company based on its current Net Asset Value plus the Present Value of the after-tax cash flows from its *current* policies.
- **M&A and Merger Models:** There are many differences; banks tend to use only Debt or Stock to fund deals, there are mark-to-market Balance Sheet adjustments, Core Deposit Intangibles get created, the Seller's Allowance for Loan Losses is written down, and there may be Deposit Divestitures; Cost Savings and Restructuring are very important; the Capital Ratios, Dividends, and Federal Funds line items will change; you may look beyond EPS accretion/dilution and assess deals with IRR, the NPV of Synergies, and other methods.
- **Buyout and Growth Equity Models:** Traditional leveraged buyouts are rare for commercial banks (more common in insurance and other sectors of financial services); PE firms typically make minority-stake investments, purchase specific assets, or do club deals if they invest in the sector; returns sources are TBV growth, P / TBV multiple expansion, and Dividends; banks can rarely take on additional Debt to fund a deal.

[Return to Top.](#)